

Using Force to Derive Maxwell's Equations

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Abstract

A derivation of Maxwell's equations is presented. We begin with the relativistic equation of motion and introduce the force tensor $R_{\mu\nu}$ with which the four-force K_μ can be written as $K_\mu = u_\nu R_{\mu\nu}$, where u_μ is the four-velocity. We show that if such a tensor exists it must be anti-symmetric. Then, we show that the electrostatic force can be written with such an anti-symmetric tensor and introduce the electromagnetic tensor $F_{\mu\nu}$ such that $K_\mu = q u_\nu F_{\mu\nu}$. The Lorentz force is derived from the Lorentz transformation of $F_{\mu\nu}$. The first set of Maxwell's equations is derived from $\partial_\nu F_{\mu\nu}$. From the symmetry of the Lorentz transformation of $F_{\mu\nu}$, we introduce dual tensor $\tilde{F}_{\mu\nu}$ from which we derive the second set of Maxwell's equations. Although we have to introduce four-vector, the calculation effort required is much less than other approaches, because we separated velocity out from the four-force by introducing the force-tensor.

1 Introduction

There are a few other derivations widely available. Schwartz[1] begins with Poisson's equation and transforms it into a Lorentz covariant form by introducing the vector potential and compensating missing terms leading to the electromagnetic tensor. Purcell[2], Elliott[3] and Haskell[4] use Lorentz transformations of electrostatic equations. Purcell's treatment is the most pedagogical while Elliott is more straight forward. Both use ordinary three dimensional vectors. Haskell follows Elliott's procedure but uses four-vector notation.

In this report, we begin with the relativistic equation of motion and introduce force tensor $R_{\mu\nu}$ with which the four-force K_μ can be written as $K_\mu = u_\nu R_{\mu\nu}$, where u_μ is the four-velocity. We show that if such a tensor exists it must be anti-symmetric. Then, we demonstrate that the electrostatic force can be written with such an anti-symmetric tensor and introduce the electromagnetic tensor $F_{\mu\nu}$ such that $K_\mu = q u_\nu F_{\mu\nu}$. The Lorentz force is derived from the Lorentz transformation of $F_{\mu\nu}$. The first set of Maxwell's equations is derived from $\partial_\nu F_{\mu\nu}$. From the symmetry of the Lorentz transformation of $F_{\mu\nu}$, we introduce the dual tensor $\tilde{F}_{\mu\nu}$ from which we derive the second set of Maxwell's equations.

This derivation stands similar position as Schwartz's but without introducing potentials. It makes it more evident that the Lorentz force and the Lorentz transformation of fields originates from the force tensor in the Minkowski space. Although we have to introduce four-vector, the calculation effort required is much less than other approaches, because we separated velocity out from the four-force by introducing the force-tensor.

This report is intended for non-experts. We refrain from using advanced mathematics such as the metric tensor. We follow the notation used in the Feynman's Lectures[5], which is summarized in Appendix A.

2 Derivation

2.1 Reference Frames

We consider two inertial frames S and S' such that S' is moving at velocity $\mathbf{V}$ relative to S . We will use γ and β defined below.

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}, \quad \beta = |\boldsymbol{\beta}|, \quad \boldsymbol{\beta} = \mathbf{V}/c. \quad (2.1)$$

We will use these only for relative velocity between inertial frames, not for velocity of the particle to avoid confusion. The Lorentz transformation of a four-vector $(x_t, \mathbf{x})$ from S to S' is given below.

$$\mathbf{x}' = \mathbf{x}_\perp + \gamma (\mathbf{x}_\parallel - \boldsymbol{\beta} x_t), \quad x'_t = \gamma (x_t - \boldsymbol{\beta} \cdot \mathbf{x}), \quad (2.2)$$

where $\mathbf{x}_\perp$ and $\mathbf{x}_\parallel$ is normal and parallel component of $\mathbf{x}$ with respect to $\boldsymbol{\beta}$, namely,

$$\mathbf{x} = \mathbf{x}_\perp + \mathbf{x}_\parallel, \quad \mathbf{x}_\parallel = \frac{\boldsymbol{\beta} (\boldsymbol{\beta} \cdot \mathbf{x})}{\beta^2}. \quad (2.3)$$

The inverse transformation is obtained by inverting $\mathbf{V}$, thus $\boldsymbol{\beta}$.

$$\mathbf{x} = \mathbf{x}'_\perp + \gamma (\mathbf{x}'_\parallel + \boldsymbol{\beta} x'_t), \quad x_t = \gamma (x'_t + \boldsymbol{\beta} \cdot \mathbf{x}'). \quad (2.4)$$

2.2 Force Tensor

Let us begin with the relativistic equation of motion of a particle whose mass and coordinate are m and $(ct, \mathbf{r})$, respectively.

$$mc^2 \frac{du_\mu}{ds} = K_\mu,$$

where u_μ is the four-velocity of the particle.

$$u = u_\mu = \frac{d(ct, \mathbf{r})}{ds} = \frac{(cdt, d\mathbf{r})}{cdt\sqrt{1-(v/c)^2}} = \frac{1}{\sqrt{1-(v/c)^2}} \left(1, \frac{\mathbf{v}}{c}\right). \quad (2.5)$$

K_μ is the four-force acting on the particle.

$$K = K_\mu = \frac{1}{\sqrt{1-(v/c)^2}} \left(\frac{\mathbf{v}}{c} \cdot \mathbf{f}, \mathbf{f}\right), \quad (2.6)$$

where $\mathbf{f}$ is ordinary force.

The fact that the force K_μ depends on the velocity of the particle motivates us to explore possibilities to separate it into a velocity dependent part and a velocity independent part. Suppose that we can express four-force K_μ by four-velocity u_μ and a tensor $R_{\mu\nu}$, like below.

$$mc^2 \frac{du_\mu}{ds} = K_\mu = u_\nu R_{\mu\nu}$$

If such $R_{\mu\nu}$ exists, it must be anti-symmetric. The explanation[1] follows.

Taking inner product with u_μ yields

$$mc^2 u_\mu \frac{du_\mu}{ds} = u_\mu u_\nu R_{\mu\nu}.$$

The left hand side is zero, because $u_\mu u_\mu = 1$.

$$mc^2 u_\mu \frac{du_\mu}{ds} = \frac{mc^2}{2} \frac{du_\mu u_\mu}{ds} = 0$$

Therefore, $u_\mu u_\nu R_{\mu\nu}$ must be zero which means $R_{\mu\nu}$ must be anti-symmetric. We have summarized the notation we will use below and some basic properties of anti-symmetric tensors in Appendix A.3.

We get Eq. (2.6) with $R = R_{\mu\nu} = (\mathbf{f}, 0)$:

$$\begin{aligned} uR &= (u_t, \mathbf{u}) (\mathbf{f}, 0), \\ &= \frac{1}{\sqrt{1-(v/c)^2}} \left(\frac{\mathbf{v}}{c} \cdot \mathbf{f}, \mathbf{f}\right) = K, \end{aligned}$$

where we used Eqs. (A.2) and (2.5).

In case $\mathbf{R} = (\mathbf{p}, \mathbf{q})$,

$$\begin{aligned} \mathbf{K} = \mathbf{uR} &= (u_t, \mathbf{u})(\mathbf{p}, \mathbf{q}) = (\mathbf{u} \cdot \mathbf{p}, u_t \mathbf{p} + \mathbf{u} \times \mathbf{q}), \\ &= \frac{1}{\sqrt{1 - (v/c)^2}} \left(\frac{\mathbf{v}}{c} \cdot \mathbf{p}, \mathbf{p} + \frac{\mathbf{v}}{c} \times \mathbf{q} \right). \end{aligned}$$

Therefore,

$$\mathbf{f} = \mathbf{p} + \frac{\mathbf{v}}{c} \times \mathbf{q}.$$

In case both $\mathbf{p}$ and $\mathbf{q}$ does not depend of $\mathbf{v}$, the Lorentz transformation of $\mathbf{R} = (\mathbf{p}, \mathbf{q})$ will be, from Eqs. (A.3) and (A.4),

$$\mathbf{p}' = \mathbf{p}_{\parallel} + \gamma(\mathbf{p} + \boldsymbol{\beta} \times \mathbf{q})_{\perp} \quad (2.7)$$

$$\mathbf{q}' = \mathbf{q}_{\parallel} + \gamma(\mathbf{q} - \boldsymbol{\beta} \times \mathbf{p})_{\perp} \quad (2.8)$$

The inverse transformation of above is obtained by changing the sign of $\boldsymbol{\beta}$.

$$\mathbf{p} = \mathbf{p}'_{\parallel} + \gamma(\mathbf{p}' - \boldsymbol{\beta} \times \mathbf{q}')_{\perp}, \quad (2.9)$$

$$\mathbf{q} = \mathbf{q}'_{\parallel} + \gamma(\mathbf{q}' + \boldsymbol{\beta} \times \mathbf{p}')_{\perp}. \quad (2.10)$$

2.3 Maxwell's Equations

We assume continuity of charge holds in any inertial frame,

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{j} = 0,$$

where ρ and $\mathbf{j}$ are charge density and current density, respectively. This leads to ρ and $\mathbf{j}$ forming a four-vector $\mathbf{j} = (c\rho, \mathbf{j})$, because this continuity equation can be viewed as the inner product of the four-gradient and the four-current is zero, or a scalar.

Suppose that all the charge is at rest in S' , i.e., $\mathbf{j}' = 0$.

$$\mathbf{j}' = (c\rho', \mathbf{j}'), \quad \mathbf{j}' = 0. \quad (2.11)$$

Static charge density ρ' creates static electric field $\mathbf{E}'$.

$$\boldsymbol{\nabla}' \cdot \mathbf{E}' = \rho' / \varepsilon_0, \quad \boldsymbol{\nabla}' \times \mathbf{E}' = 0 \quad (2.12)$$

Let us consider motion of a charged particle whose charge q is very small so that its motion will not affect the rest of charges and thus the charge density. We assume q is Lorentz invariant, i.e., $q' = q$. The force acting on the charged particle $\mathbf{f}'$ is

$$\mathbf{f}' = q\mathbf{E}'.$$

Since $\mathbf{E}'$ does not depend on the velocity $\mathbf{v}$ of the particle, we can use the force tensor.

$$\mathbf{K}' = q(u'_t, \mathbf{u}')(\mathbf{E}', 0).$$

In S , all charges except for q are moving uniformly at $\mathbf{V}$, four-force $\mathbf{K}$ is, according to Eqs. (2.9) and (2.10),

$$\mathbf{K} = q(u_t, \mathbf{u})(\mathbf{E}'_{\parallel} + \gamma \mathbf{E}'_{\perp}, \boldsymbol{\beta} \times \gamma \mathbf{E}').$$

This gives

$$\mathbf{f} = q\left(\mathbf{E}'_{\parallel} + \gamma \mathbf{E}'_{\perp} + \frac{\mathbf{v}}{c} \times (\boldsymbol{\beta} \times \gamma \mathbf{E}')\right).$$

$(\mathbf{E}'_{\parallel} + \gamma \mathbf{E}'_{\perp})$ is electric field in S and $(\boldsymbol{\beta} \times \gamma \mathbf{E}')$ can be interpreted as magnetic field.

Now, we define the electromagnetic tensor $\mathbf{F}$ as below.¹

$$c\mathbf{F} = (\mathbf{E}, c\mathbf{B})$$

Giving four-velocity of the particle $\mathbf{u}$ to $\mathbf{F}$ yields four-force $\mathbf{K}$.

$$\begin{aligned} q \mathbf{u} c\mathbf{F} &= \mathbf{K} = q(u_t, \mathbf{u})(\mathbf{E}, c\mathbf{B}) = q(\mathbf{u} \cdot \mathbf{E}, u_t \mathbf{E} + \mathbf{u} \times c\mathbf{B}), \\ &= \frac{q}{\sqrt{1 - (v/c)^2}} \left(\frac{\mathbf{v}}{c} \cdot \mathbf{E}, \mathbf{E} + \frac{\mathbf{v}}{c} \times c\mathbf{B} \right) \end{aligned}$$

Applying the four-gradient ∂ to $\mathbf{F}$ yields a four-vector.

$$\partial c\mathbf{F} = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\boldsymbol{\nabla} \right) (\mathbf{E}, c\mathbf{B}) = \left(-\boldsymbol{\nabla} \cdot \mathbf{E}, \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} - \boldsymbol{\nabla} \times c\mathbf{B} \right) \quad (2.13)$$

In S' where $c\mathbf{F}' = (\mathbf{E}', 0)$, $\mathbf{E}'$ is static and $\boldsymbol{\nabla}' \cdot \mathbf{E}' = \rho'/\epsilon_0$,

$$\partial' c\mathbf{F}' = \left(\frac{1}{c} \frac{\partial}{\partial t'}, -\boldsymbol{\nabla}' \right) (\mathbf{E}', 0) = (-\boldsymbol{\nabla}' \cdot \mathbf{E}', 0) = (\rho'/\epsilon_0, 0).$$

We see that the time component of Eq. (2.13) should be $-\rho/\epsilon_0$ and the spatial component must be equal to $-\mathbf{j}/\epsilon_0 c$ to make Eq. (2.13) a four-vector. Therefore,

$$\partial c\mathbf{F} = -\mathbf{j}/\epsilon_0 c, \quad \text{or} \quad \partial_{\nu} cF_{\mu\nu} = -j_{\mu}/\epsilon_0 c,$$

or in ordinary vector,

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \boldsymbol{\nabla} \times c\mathbf{B} - \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} = \frac{\mathbf{j}}{\epsilon_0 c}. \quad (2.14)$$

¹We follow SI units. It will be $\mathbf{B}$ instead of $c\mathbf{B}$ in Gaussian units. See Ref. [6] for details.

$\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ holds not only for static field but also for $\mathbf{E}$ created by moving charge.

The Lorentz transformation of $\mathbf{E}$ and $c\mathbf{B}$ is, according to Eqs. (2.7) and (2.8),

$$\mathbf{E}' = \mathbf{E}_{\parallel} + \gamma(\mathbf{E} + \boldsymbol{\beta} \times c\mathbf{B})_{\perp}, \quad (2.15)$$

$$c\mathbf{B}' = c\mathbf{B}_{\parallel} + \gamma(c\mathbf{B} - \boldsymbol{\beta} \times \mathbf{E})_{\perp}. \quad (2.16)$$

We see symmetry between $\mathbf{E}$ and $c\mathbf{B}$ in above Lorentz transformation. The Lorentz transformation remains the same under following substitution.

$$\mathbf{E} \rightarrow c\mathbf{B}, \quad c\mathbf{B} \rightarrow -\mathbf{E},$$

Therefore, we have another anti-symmetric tensor, which gives the same Lorentz transformation for $\mathbf{E}$ and $c\mathbf{B}$. That is

$$c\tilde{\mathbf{F}} = (c\mathbf{B}, -\mathbf{E}).$$

Applying four-gradient ∂ to $\tilde{\mathbf{F}}$ yields a four-vector.

$$\partial c\tilde{\mathbf{F}} = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\nabla \right) (c\mathbf{B}, -\mathbf{E}) = \left(-\nabla \cdot c\mathbf{B}, \frac{1}{c} \frac{\partial c\mathbf{B}}{\partial t} + \nabla \times \mathbf{E} \right)$$

In S' , we don't have any ordinary force component that depends on the velocity of the particle, therefore $c\mathbf{B}' = 0$. $\mathbf{E}'$ is static, $\nabla' \times \mathbf{E}' = 0$. Therefore, this vector is zero in S' . Because the Lorentz transformation is a linear transformation, this vector must be zero in any inertial frame. Therefore

$$\partial c\tilde{\mathbf{F}} = 0, \quad \text{or} \quad \partial_{\nu} c\tilde{F}_{\mu\nu} = 0,$$

or in ordinary vector,

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial c\mathbf{B}}{\partial t} = 0. \quad (2.17)$$

Eqs. (2.14) and (2.17) are the Maxwell's equations.

3 Concluding Remarks

We made following assumptions:

- (a) Charge of the particle is Lorentz invariant. That is $q' = q$.
- (b) Continuity of charge holds any inertial frame.
- (c) $\mathbf{E}'$ is static. $\nabla' \cdot \mathbf{E}' = \rho'/\epsilon_0$, $\nabla' \times \mathbf{E}' = 0$, and $\mathbf{j}' = (c\rho', 0)$.

(d) Force $\mathbf{f}'$ acting on moving charged particle in static electric field $\mathbf{E}'$ is $\mathbf{f}' = q\mathbf{E}'$.

(b) can be derived from (a) if we employ following charged particle model.

$$\rho = \sum_i q_i \delta^3(\mathbf{r} - \mathbf{r}_i), \quad \mathbf{j} = \sum_i \mathbf{v}_i q_i \delta^3(\mathbf{r} - \mathbf{r}_i).$$

(c) is consequence of Coulomb's law (including principle of superposition). (d) is subject to verification, i.e., we need to ask nature if this is true (and she says yes).

Note that the derivation presented here is just a derivation, not a proof. We have demonstrated that Maxwell's equations can be obtained from special relativity and Coulomb's law and the invariance of charge. But we did not prove that Maxwell's equations are the only possibility nor guarantee that they are applicable beyond uniformly moving charges. They are still subject to verification. And yes, we all know that Maxwell's equations work very well to describe reality.

Appendix A Vectors and tensors

This report is intended for non-experts. We refrain from using advanced mathematics such as metric tensor. We follow the notation used in the Feynman's Lectures[5].

A.1 Vectors

We use following notation for ordinary vectors. We use $i, j, k, \dots$ for index x, y, z .

$$\mathbf{a} = a_i = (a_x \quad a_y \quad a_z)$$

Below is inner product of ordinary vectors. Summation rule applies.

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i = a_x b_x + a_y b_y + a_z b_z$$

We use following notation for four-vector. We use $\mu, \nu, \rho, \dots$ for index t, x, y, z .

$$\mathbf{a} = (a_t, \mathbf{a}) = a_\mu = (a_t \quad a_x \quad a_y \quad a_z)$$

Below is inner product of four-vectors. Summation rule applies.

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= (a_t, \mathbf{a}) \cdot (b_t, \mathbf{b}), \\ &= a_\mu b_\mu, \\ &= a_t b_t - \mathbf{a} \cdot \mathbf{b}, \\ &= a_t b_t - a_x b_x - a_y b_y - a_z b_z \end{aligned}$$

A.2 Four-gradient

Suppose a scalar function f of t and $\mathbf{r}$, and recalling that the total differential df of a scalar function $f(t, \mathbf{r})$ is also a scalar.

$$df = \frac{\partial f}{\partial t} dt + \nabla f \cdot d\mathbf{r}$$

This can be viewed as inner product of a displacement four-vector $d\mathbf{r} = (c dt, d\mathbf{r})$ and the four-gradient ∂ .

$$df = \partial f \cdot d\mathbf{r} \quad \text{where} \quad \partial = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\nabla \right)$$

Similarly, continuity of charge,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0,$$

can be viewed as inner product of the four-gradient ∂ and the four-current $\mathbf{j}$.

$$\partial \cdot \mathbf{j} = 0 \quad \text{where} \quad \mathbf{j} = (c\rho, \mathbf{j})$$

A.3 Anti-symmetric tensor

An anti-symmetric tensor is a second-rank tensor whose component flips its sign when its index order is flipped, like below.

$$A_{\mu\nu} = -A_{\nu\mu}$$

It has six independent components.

$$A_{\mu\nu} = \begin{pmatrix} 0 & A_{tx} & A_{ty} & A_{tz} \\ A_{xt} & 0 & A_{xy} & A_{xz} \\ A_{yt} & A_{yx} & 0 & A_{yz} \\ A_{zt} & A_{zx} & A_{zy} & 0 \end{pmatrix} = \begin{pmatrix} 0 & -A_{xt} & -A_{yt} & -A_{zt} \\ A_{xt} & 0 & -A_{yx} & A_{xz} \\ A_{yt} & A_{yx} & 0 & -A_{zy} \\ A_{zt} & -A_{xz} & A_{zy} & 0 \end{pmatrix}$$

Let us work on the case when it is written with two four-vectors as below.

$$A_{\mu\nu} = a_\mu b_\nu - a_\nu b_\mu$$

Because any anti-symmetric tensor can be written with a linear combination of these, extending below discussion to general case is straight forward.

A.3.1 Two ordinary vectors

$\mathbf{p} = (A_{xt} \ A_{yt} \ A_{zt})$ forms an ordinary vector because it is linear combination of two ordinary vectors (or linear combination of those).

$$\begin{aligned}\mathbf{p} &= (A_{xt} \ A_{yt} \ A_{zt}) \\ &= (a_x b_t - a_t b_x \ a_y b_t - a_t b_y \ a_z b_t - a_t b_z), \\ &= (a_x \ a_y \ a_z) b_t - a_t (b_x \ b_y \ b_z), \\ &= \mathbf{a} b_t - a_t \mathbf{b}\end{aligned}$$

$\mathbf{q} = (A_{zy} \ A_{xz} \ A_{yx})$ forms an ordinary vector too, because it is vector product of two ordinary vectors (or linear combination of those).

$$\begin{aligned}\mathbf{q} &= (A_{zy} \ A_{xz} \ A_{yx}), \\ &= ((a_z b_y - a_y b_z) \ (a_x b_z - a_z b_x) \ (a_y b_x - a_x b_y)), \\ &= -\mathbf{a} \times \mathbf{b}\end{aligned}$$

We use $(\mathbf{p}, \mathbf{q})$ to represent $\mathbf{A}$.

$$\mathbf{A} = (\mathbf{p}, \mathbf{q}) = A_{\mu\nu} = \begin{pmatrix} 0 & -p_x & -p_y & -p_z \\ p_x & 0 & -q_z & q_y \\ p_y & q_z & 0 & -q_x \\ p_z & -q_y & q_x & 0 \end{pmatrix} \quad (\text{A.1})$$

When both $\mathbf{a}$ and $\mathbf{b}$ are polar, $\mathbf{p}$ is polar and $\mathbf{q}$ is axial.

A.3.2 Linear map

Giving four-vector w to anti-symmetric tensor $\mathbf{A}$ yields a four-vector.

$$\begin{aligned}w\mathbf{A} &= (w_t, \mathbf{w}) (\mathbf{p}, \mathbf{q}) = w_\nu A_{\mu\nu} \\ &= w_t A_{\mu t} - w_x A_{\mu x} - w_y A_{\mu y} - w_z A_{\mu z}.\end{aligned}$$

t, x, y, z components are, respectively,

$$\begin{aligned}w_\nu A_{t\nu} &= w_t A_{tt} - w_x A_{tx} - w_y A_{ty} - w_z A_{tz}, \\ &= w_x p_x + w_y p_y + w_z p_z = \mathbf{w} \cdot \mathbf{p}, \\ w_\nu A_{x\nu} &= w_t A_{xt} - w_x A_{xx} - w_y A_{xy} - w_z A_{xz}, \\ &= w_t p_x + w_y q_z - w_z q_y, \\ w_\nu A_{y\nu} &= w_t A_{yt} - w_x A_{yx} - w_y A_{yy} - w_z A_{yz}, \\ &= w_t p_y + w_z q_x - w_x q_z, \\ w_\nu A_{z\nu} &= w_t A_{zt} - w_x A_{zx} - w_y A_{zy} - w_z A_{zz}, \\ &= w_t p_z + w_x q_y - w_y q_x.\end{aligned}$$

Therefore

$$\mathbf{w}\mathbf{A} = (w_t, \mathbf{w}) (\mathbf{p}, \mathbf{q}) = w_\nu A_{\mu\nu} = (\mathbf{w} \cdot \mathbf{p}, w_t \mathbf{p} + \mathbf{w} \times \mathbf{q}). \quad (\text{A.2})$$

A.3.3 Lorentz transformation

Let us use Cartesian coordinates whose x-axis coincides with $\boldsymbol{\beta}$, i.e., $\boldsymbol{\beta} = (\beta_x \ 0 \ 0)$. The Lorentz transformation of $\mathbf{a}$ and $\mathbf{b}$ is, respectively,

$$a'_x = \gamma (a_x - \beta_x a_t), \quad a'_y = a_y, \quad a'_z = a_z, \quad a'_t = \gamma (a_t - \beta_x a_x),$$

and

$$b'_x = \gamma (b_x - \beta_x b_t), \quad b'_y = b_y, \quad b'_z = b_z, \quad b'_t = \gamma (b_t - \beta_x b_x).$$

For $\mathbf{p} = (A_{xt} \ A_{yt} \ A_{zt})$,

$$\begin{aligned} p'_x &= A'_{xt} = a'_x b'_t - a'_t b'_x, \\ &= \gamma^2 (a_x - \beta_x a_t) (b_t - \beta_x b_x) - \gamma^2 (a_t - \beta_x a_x) (b_x - \beta_x b_t), \\ &= \gamma^2 (1 - \beta_x^2) a_x b_t - \gamma^2 (1 - \beta_x^2) a_t b_x = a_x b_t - a_t b_x \\ &= A_{xt} = p_x, \\ p'_y &= A'_{yt} = a'_y b'_t - a'_t b'_y, \\ &= \gamma a_y (b_t - \beta_x b_x) - \gamma (a_t - \beta_x a_x) b_y, \\ &= \gamma (a_y b_t - a_t b_y - \beta_x a_y b_x + \beta_x a_x b_y) \\ &= \gamma (A_{yt} - \beta_x A_{yx}) = \gamma (p_y - \beta_x q_z), \\ p'_z &= A'_{zt} = \gamma (A_{zt} + \beta_x A_{xz}) = \gamma (p_z + \beta_x q_y). \end{aligned}$$

Similarly, for $\mathbf{q} = (A_{zy} \ A_{xz} \ A_{yx})$,

$$\begin{aligned} q'_x &= A'_{zy} = A_{zy} = q_x, \\ q'_y &= A'_{xz} = \gamma (A_{xz} + \beta_x A_{zt}) = \gamma (q_y + \beta_x p_z), \\ q'_z &= A'_{yx} = \gamma (A_{yx} - \beta_x A_{yt}) = \gamma (q_z - \beta_x p_y). \end{aligned}$$

To convert these to the vector notation, we need to recover missing terms. For example, when z component has term $\beta_x q_y$, we need to add $-\beta_y q_x$ to make it a vector. Then,

$$\mathbf{p}'_{\parallel} = \mathbf{p}_{\parallel}, \quad \mathbf{p}'_{\perp} = \gamma (\mathbf{p} + \boldsymbol{\beta} \times \mathbf{q})_{\perp}, \quad (\text{A.3})$$

$$\mathbf{q}'_{\parallel} = \mathbf{q}_{\parallel}, \quad \mathbf{q}'_{\perp} = \gamma (\mathbf{q} - \boldsymbol{\beta} \times \mathbf{p})_{\perp}. \quad (\text{A.4})$$

The inverse transformation can be obtained by inverting the sign of $\boldsymbol{\beta}$.

$$\mathbf{p}_{\parallel} = \mathbf{p}'_{\parallel}, \quad \mathbf{p}_{\perp} = \gamma (\mathbf{p}' - \boldsymbol{\beta} \times \mathbf{q}')_{\perp}, \quad (\text{A.5})$$

$$\mathbf{q}_{\parallel} = \mathbf{q}'_{\parallel}, \quad \mathbf{q}_{\perp} = \gamma (\mathbf{q}' + \boldsymbol{\beta} \times \mathbf{p}')_{\perp}. \quad (\text{A.6})$$

A.3.4 Dual tensor

We see symmetry between $\mathbf{p}$ and $\mathbf{q}$ in their Lorentz transformation (Eqs. (A.3), (A.4)), i.e., following substitution will give the same Lorentz transformation.

$$\mathbf{p} \rightarrow \mathbf{q}, \quad \mathbf{q} \rightarrow -\mathbf{p}.$$

Therefore, $\tilde{\mathbf{A}} = (\mathbf{q}, -\mathbf{p})$ gives the same transformation for $\mathbf{p}$ and $\mathbf{q}$ as $\mathbf{A} = (\mathbf{p}, \mathbf{q})$.

$$\tilde{\mathbf{A}} = (\mathbf{q}, -\mathbf{p}) = \tilde{A}_{\mu\nu} = \begin{pmatrix} 0 & -q_x & -q_y & -q_z \\ q_x & 0 & p_z & -p_y \\ q_y & -p_z & 0 & p_x \\ q_z & p_y & -p_x & 0 \end{pmatrix} \quad (\text{A.7})$$

Dual tensor is valid only under “proper” (not inverting) coordinate transformations, since we are substituting polar vector with axial vector, vice versa.

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